

Question			Solution	Marks
4	(a)	(i)	$\text{e.m.f.} = (\Delta)B \times AN / t$ $= 45 \times 10^{-3} \times \pi \times (1.8 \times 10^{-2})^2 \times 350 / 0.20 = 0.080 \text{ V}$	C1 A1
		(ii)	0 to 0.2 s: straight horizontal line at 0.080 V or –0.080 V 0.2 s to 0.4 s: zero 0.4 s to 0.8 s: straight horizontal line at 0.040 V or –0.040 V opposite polarity to 0 to 0.2 s line	B1 B1 B1 B1
	(b)		either disc cuts flux lines (of the magnet) or there is a changing flux in the disc (by Faraday's law) e.m.f. is induced in the disc e.m.f. causes (eddy) currents in the disc current in the disc causes force on disc (accepts FLHR)	B1 B1 B1 B1

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5	(a)	$n = 110 / 0.032$ or $110000 / 32$ or 3440	C1